AI 500 - Foundations of AI - ANSWER KEY
Quiz 3 - Unsupervised Learning - 5th October 2025

Name:_____ Roll Number:_____

Please note that MCQs can have multiple correct options.

Q1. What is true for K-means? **(2 marks)**

A. It requires the number of clusters k as input.
B. It minimises the sum of squared distances to cluster *medoids*.
C. It is sensitive to initialization.
D. It assumes roughly spherical clusters under Euclidean distance.

Q2. What is the difference between K-medoid and K-means? **(2 marks)**

A. K-medoid uses actual data points as centers.
B. K-medoid is generally more robust to outliers than K-means.
C. K-means centers minimize L1 error, while K-medoid minimizes L2 error.
D. Replacing means by medoids can alleviate outlier sensitivity.

Q3. When would you prefer DBSCAN over K-means? **(1 mark)**

When clusters are of arbitrary shape, density-separable, and when you need explicit handling of noise/outliers without pre-specifying k.

You have been given the following data:

- X=[2,5,6,7,8,9,110]
- k=2
- Initial centroids c1 = 5, c2 = 9

Q4. Run one iteration of K-means algorithm to do the following:

a) Assign each point to its nearest centroid. **(2 marks)**
b) Compute updated centroids. **(2 marks)**

Distances to c1=5:=3,0,1,2,3,4,105.
Distances to c2=9:|=7,4,3,2,1,0,101.
Assignments :
Cluster 1: [2,5,6,7]
Cluster 2: [8,9,110]

Update:
Cluster 1 centroid = (2+5+6+7)/4=5
Cluster 2 centroid = (8+9+10)/3 = 42.33

You have been given points from P1-P5. Run DBSCAN with ε = 1.1 and MinPts = 3 (counting the point itself). You have been given the following pairwise distances**:**

- d(P1,P2)=1.0
- d(P1,P3)=1.0
- d(P4,P5)=1.0

All other pairwise distances are $> \varepsilon$. Assume P1 is already a core point. Using this information:

a) Classify P2, P3, P4, and P5 as Core / Border / Noise. **(4 marks)**
b) List the final cluster(s) and noise set. **(1 mark)**

P2:[P2,P1] - 2<3 → not core; it is within ε of core P1 → Border.
P3:[P3,P1] - 2<3 → not core; within ε of core P1 → Border.
P4:[P4,P5] - 2<3 → not core; not within ε of any core point → Noise.
P5:[P5,P4] - 2<3 → not core; not within ε of any core point → Noise.
Clusters: [P1,P2,P3]
Noise: [P4,P5]